

Code-Layout, Readability and Reusability

Date	07 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Files are organized into templates, models, and application folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables clearly represent applicant and loan details.
4	Function / Method Names	Functions are descriptively named	Partial	Functions like predict() and submit() are descriptive.
5	Code Comments	Inline and block comments explain log	Partial	Important sections are commented for readability.
6	Modular Design	Code is split into reusable functions/modules	Yes	Model, scaler, and Flask application are separated into modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code is minimized.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and prediction handling are implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Backend Routing	High
2	Random Forest Model (model.pkl)	Scikit-learn	Loan Prediction	High
3	Standard Scaler (scaler.pkl)	Scikit-learn	Data Preprocessing	High
4	HTML Templates	HTML, CSS	User Interface	Medium
5	Prediction Function	Python	Prediction Module	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized project structure.
Readability	5	Easy-to-understand code and naming.
Reusability	5	Model, scaler, and templates can be reused.
Documentation / Comments	4	Basic comments are provided.
Overall Score	5	Clean, modular, and maintainable implementation